CORPORATE INCOME TAX RATES

Learning Outcome

- calculate, interpret, and contrast an issuer's effective tax rate, statutory tax rate, and cash tax rate

Income taxes payable are primarily determined by the geographic composition of taxable income and the tax rates in each jurisdiction but can also be influenced by the nature of a business. Some companies benefit from special tax treatment—for example, from R&D tax credits or accelerated depreciation of fixed assets. Analysts should also be aware of any governmental or business changes that can alter tax rates.

Differences in tax rates can be an important driver of value. Generally, three types of tax rates are relevant to analysts:

- The **statutory tax rate**, which is the corporate income tax rate in the country in which the company is domiciled.
- The **effective tax rate**, which is calculated as the reported income tax expense amount on the income statement divided by the pre-tax income.
- The **cash tax rate**, which is the tax paid in cash that period (cash tax) divided by pre-tax income.

As discussed previously, differences between cash taxes and reported taxes typically result from differences between financial accounting standards and tax laws and result from changes in deferred tax assets or deferred tax liabilities.

In forecasting tax expense and cash taxes, respectively, the effective tax rate and cash tax rate are key. A good understanding of their operational drivers and the financial structure of a company is useful in forecasting these tax rates.

Differences between the statutory tax rate and the effective tax rate can arise for many reasons. Tax credits, withholding tax on dividends, adjustments to previous years, and expenses not deductible for tax purposes are among the reasons for differences. Effective tax rates can also differ when companies are active outside the country in which they are domiciled. The effective tax rate becomes a blend of the different tax rates of the countries in which the activities take place in relation to the profit generated in each country. If a company reports a high profit in a country with a high tax rate and a low profit in a country with a low tax rate, the effective tax rate will be the weighted average of the rates and higher than the simple average tax rate of both countries.

In general, an effective tax rate that is consistently lower than statutory rates or the effective tax rates reported by competitors is not necessarily unusual but might warrant additional attention when forecasting future tax expenses. The notes in the financial statements should disclose a reconciliation of the statutory tax rate to the effective rate and identify the items that significantly contribute to a temporarily high or low effective tax rate. The cash tax rate is used for forecasting cash flows, and the effective tax rate is relevant for projecting earnings on the income statement.

In developing an estimated tax rate for forecasts, analysts should adjust for one-time events. If the income from equity-method investees is a substantial part of pre-tax income and, also a volatile component of it, the effective tax rate excluding this amount is likely to be a better estimate for the future tax costs for a company. The tax impact from income from participations is disclosed in the notes on the financial statements.

normalized tax rate should be a good indication of the future tax expense, adjusted for special items, in an analyst's earnings model.

Building a model allows the effective tax amount to be found in the profit and loss projections and the cash tax amount on the cash flow statement (or given as supplemental information). The reconciliation between the profit and loss tax amount and the cash flow tax figures should be the change in the deferred tax asset or liability.

EXAMPLE 4

Tax Rate Estimates

ABC, a hypothetical company, operates in Countries A and B. The tax rate in Country A is 40 percent, and the tax rate in Country B is 10 percent. In the first year, the company generates an equal amount of profit before tax in each country, as shown in Exhibit 11.

Exhibit 11: Tax Rates That Differ by Jurisdiction

	A	B	Total
Profit before tax	100	100	200
Effective tax rate	40%	10%	25%
Tax	40	10	50
Net profit	60	90	150

1. What will happen to the effective tax rate for the next three years if the profit before tax in Country A is stable but the profit before tax in Country B grows 15 percent annually?

Hide Solution

Solution:

The effective tax rate will gradually decline because a higher proportion of profit will be generated in the country with the lower tax rate each year. In Exhibit 12, the effective tax rate declines from 25 percent in the beginning to 22 percent in the third year.

Exhibit 12: Worksheet for Tax Rate Estimates Problem

	Year			
	0	1	2	3
Profit before tax, Country A	100	100	100	100
Growth rate		0%	0%	0%
Profit before tax, Country B	100	115	132	152
Growth rate		15%	15%	15%
Total profit before tax	200	215	232	252
Effective tax rate, Country A	40%	40%	40%	40%
Effective tax rate, Country B	10%	10%	10%	10%
Total tax	50	52	53	55
Total effective tax rate	25%	24%	23%	22%

taken sooner for tax purposes. Specifically, assume for Country A, the result is a 50 percent reduction in taxes paid in the current year (Year 0) but an increase in taxes paid by the same amount in the following year (Year 1) and in subsequent years. Assume stable profit before tax in Country A and 15 percent annual before-tax-profit growth in Country B.

Hide Solution

Solution:

The combined cash tax rate (last line in Exhibit 13) will be 15 percent in the first year and then rebound in subsequent years. Only the rate for the first year will benefit from a tax deferral; in subsequent years, the deferral for a given year will be offset by the addition of the amount postponed from the previous year. The combined effective tax rate will be unaffected by the deferral. As shown in Exhibit 13, beginning with the second year (Year 1), the combined cash tax and effective tax rates decline over time because the growth in taxable income occurs in Country B, which has the lower tax rate.

Exhibit 13: Worksheet for Tax Rate Estimates Problem

	Year			
	0	1	2	3
Profit before tax, Country A	100	100	100	100
Growth rate		0%	0%	0%
Profit before tax, Country B	100	115	132	152
Growth rate		15%	15%	15%
Total profit before tax	200	215	232	252
Effective tax rate, Country A	40%	40%	40%	40%
Effective tax rate, Country B	10%	10%	10%	10%
Total tax per income statement	50	52	53	55
Total effective tax rate	25%	24%	23%	22%
Cash taxes, Country A	20	40	40	40
Cash taxes, Country B	10	12	13	15
Total cash tax	30	52	53	55
Cash tax rate	15%	24%	23%	22%

3. Repeat the exercise of the Tax Rate Estimates Problem, but now assume that Country B, rather than Country A, allows some costs to be taken sooner for tax purposes and that the tax effect described applies to Country B. Continue to assume stable profit before tax in Country A and 15 percent annual profit growth in Country B.

Hide Solution

Solution:

The combined effective tax rate is unchanged from Exhibit 12 and Exhibit 13. Because of the growth assumed for Country B, however, the annual tax postponement will result in a lower cash tax rate in Country B than the effective tax rate in Country B. Consequently, as shown in Exhibit 14, the combined cash tax rate will be less than the effective tax rate in Year 0.

	Year			
	0	1	2	3
Profit before tax, Country A	100	100	100	100
Growth rate		0%	0%	0%
Profit before tax, Country B	100	115	132	152
Growth rate		15%	15%	15%
Total profit before tax	200	215	232	252
Effective tax rate, Country A	40%	40%	40%	40%
Effective tax rate, Country B	10%	10%	10%	10%
Total tax per income statement	50	52	53	55
Total effective tax rate	25%	24%	23%	22%
Cash taxes, Country A	40	40	40	40
Cash taxes, Country B	5	11	12	14
Total cash tax	45	51	52	54
Cash tax rate	23%	24%	23%	22%

EXAMPLE 5

Johnson & Johnson

The difference between the effective tax rate and the cash tax rate for Johnson & Johnson (JNJ) is shown in Exhibit 15.

Exhibit 15:

Johnson & Johnson Consolidated Statement of Earnings (US dollar millions)

	2021	2020	2019
Sales to customers	93,775	82,584	82,059
Cost of products sold	29,855	28,427	27,556
Gross profit	63,920	54,157	54,503
Selling, marketing, and administrative expense	24,659	22,084	22,178
Research and development expense	14,714	12,159	11,355
In-process research and development	900	181	890
Interest income	(53)	(111)	(357)
Interest expense	183	201	318
Other (income) expense, net	489	2,899	2,525
Restructuring	252	247	266
Income before income taxes	22,776	16,497	17,328
Provision for income taxes	1,898	1,783	2,209
Net income	20,878	14,714	15,119

JNJ's income tax expense for 2021 was USD1,898 million. Accordingly, JNJ's effective tax rate was 8.3 percent (USD1,898/USD22,776). This is substantially lower than JNJ's statutory tax rate of 21 percent as a US corporation. JNJ's reconciliation of its statutory tax rate to its effective tax rate reported in its notes to financial statements shows that the primary driver of this differences is a lower tax rate on its business outside the United States.

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US statutory rate	21.0%	21.0%	21.0%
International operations	(16.4)	(9.9)	(5.9)
US taxes on international income	6.7	2.7	1.8
Tax benefits from loss on capital assets	(1.3)	(1.2)	(0.3)
Tax benefits on share-based compensation	(1.0)	(1.5)	(0.5)
Tax Cuts And Jobs Act related impacts	(0.5)	0.7	(3.9)
All Other	(0.2)	(1.0)	0.5
Effective tax rate	8.3%	10.8%	12.7%

Question Set

1. Which of the following *best* describes a statutory tax rate?

A. Tax paid in cash that period divided by pre-tax income
B. Corporate income tax rate in the country in which the company is domiciled
C. Reported income tax expense amount on the income statement divided by the pre-tax income

Hide Solution

Solution:

B is correct. The statutory tax rate is the corporate income tax rate in the country in which the company is domiciled. A is incorrect because it describes the cash tax rate. C is incorrect because it describes the effective tax rate.

Please use the following information and Exhibit 16 and Exhibit 17 from Walmart's 2021 Form 10-K to answer questions 2 and 3:

Walmart reported total revenues of USD572,762 million, income before taxes of USD18,696 million and consolidated net income of USD13,940 million. In addition, it disclosed the following information in its tax footnote:

Exhibit 16: Walmart's 2021 Form 10-K

Note 9: Taxes

The components of income before income taxes are as follows:

	Fiscal Years Ended 31 January		
(in millions USD)	**2022**	**2021**	**2020**
US	15,536	18,068	15,019
Non-US	3,160	2,496	5,097
Total income before income taxes	18,696	20,564	20,116

A summary of the provision for income taxes is as follows:

Exhibit 17: Walmart's 2021 Form 10-K

	Fiscal Years Ended 31 January		
(in millions USD)	**2022**	**2021**	**2020**
Current:			
US Federal	3,313	2,991	2,794
US State and Local	649	742	587

International	1,553	1,127	1,205
Total current tax provision	5,515	4,860	4,586
Deferred:			
US Federal	(671)	2,316	663
US State and Local	41	23	35
International	(129)	(341)	(369)
Total deferred tax expense (benefit)	(759)	1,998	329
Total provision for income taxes	4,756	6,858	4,915

2. Based on the information in Exhibit 16 and Exhibit 17, Walmart's effective tax rate is *closest* to:

A. 25.4 percent.
B. 29.5 percent.
C. 34.1 percent.

Hide Solution

Solution:

A is correct. The effective tax rate is calculated as the total provision for income taxes (USD4,756 million) divided by Total income before taxes (USD18,696 million).

3. Based on the information in Exhibit 16 and Exhibit 17, Walmart's cash tax rate is *closest* to:

A. 25.4 percent.
B. 29.5 percent.
C. 34.1 percent.

Hide Solution

Solution:

B is correct. The cash tax rate is calculated as the current tax expense (USD5,515 million) divided by Total income before taxes (USD18,696 million).

Please use the following information to answer questions 4 and 5 below:

Neutrino is a hypothetical company that is domiciled in the United States and that has significant operations in Ireland. The Statutory Tax rate in the United States is 21 percent, and the statutory tax rate in Ireland is 12 percent. Assume that Neutrino earns USD1,000 in profit before taxes in each country during year 20X1.

4. Assuming that there are no other differences between Neutrino's effective and statutory tax rates, Neutrino's combined effective tax rate is *closest* to:

A. 12.0 percent.
B. 16.5 percent.
C. 21.0 percent.

Hide Solution

Solution:

B is correct. Taxes are calculated as: (USD1,000 × 21%) + (USD1,000 × 12%) = USD330; Effective tax rate = USD330/USD2,000 = 16.5%.

5. Assume on January 1 the following year, 20X2, Neutrino acquires company EFG, which is domiciled in South Korea. The statutory tax rate in South Korea is 25 percent. EFG earns USD500 in profits in 20X2. Assuming US and Ireland operations each increase pre-tax profits by 25 percent, the effective tax rate in 20X2 for the consolidated entity is *closest* to:

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C. 22.0 percent.

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Solution:

B is correct. The effective tax rate in 20X2 for the consolidated entity is calculated as follows:

Country	Taxable Income	Statutory Rate	Taxes
United States	USD1,250	21%	USD262.50
Ireland	1,250	12	150.00
South Korea	500	25	125.00
Total	USD3,000		537.50
Effective tax rate:		537.50/3,000	**17.9%**

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SV Enigma kk uses straight line depreciation in its accounts while using accelerated depreciation for tax purposes. For both financial and tax reporting, the company assumes a five year useful economic life and zero residual value. In the current year, accounting depreciation is $10,000,000 and tax allowable depreciation is $8,000,000. Which of the following statements relating to the impact of depreciation in the current year is most accurate. The given statements are: 1) DTL will increase, 2) the tax expense will be greater than tax payable, 3) Carrying value of asset is greater than the tax base. Which of the statements is correct? I think all three are incorrect but the correct answer according to practice questions is option 3. Please can you explain how?

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BN In Ex 4, Qn 2 "Specifically, assume for Country A, the result is a 50 percent reduction in taxes paid in the current year (Year 0) but an increase in taxes paid by the same amount in the following year (Year 1) and in subsequent years. Assume stable profit before tax in Country A and 15 percent annual before-tax-profit growth in Country B."

However in the subsequent years, in Exhibit 13, which describes the answer , the Cash Tax paid is shown as 40. Shouldn't it be 60? The Deferred Tax Liability is paid off in subsequent years.

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CA Why is witholding tax mentioned in the opening segment - is this not a tax on the investor rather than the company?

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